

ROUNAK DAS

📍 Prayagraj, Uttar Pradesh 📞 +91 87774 73417 ✉️ rounakdas2025@gmail.com
🌐 az9tumas.github.io/my-portfolio 🔗 linkedin.com/in/rounak-das-600a99315 🐙 github.com/az9tumas

SUMMARY

Computer Science undergraduate at MNNIT Allahabad with 4+ years of applied experience as a lead programmer in game development. Shipped 5+ Roblox titles including games with 50M+ visits and 1,500+ concurrent players. Currently seeking software engineering internships across systems programming and data science.

TECHNICAL SKILLS

Languages	Python, C/C++, Lua
Frameworks & Libraries	NumPy, Pandas, Scikit-learn, Statsmodels (ARIMA), Matplotlib, SDL
Developer Tools	Git/GitHub, Neovim, Roblox Studio, Linux Terminal, CI/CD (GitHub Actions)
Platforms & OS	Windows, Linux, MacOS
Core Competencies	Data Structures and Algorithms (DSA), Object-Oriented Programming (OOPs), System Design, Compiler Design

PROFESSIONAL EXPERIENCE

- **Lead Programmer, Freelance Game Developer** *Roblox Platform* May 2021 – Present
 - Led end-to-end development of 5+ shipped titles, including games that reached 50M+ total visits and sustained 1,500+ concurrent players.
 - Engineered fault-tolerant datastore pipelines with rollback safeguards, maintaining data integrity and zero-loss persistence under peak concurrent load.
 - Implemented reusable, modular combat and physics frameworks covering airborne, aquatic, and ground vehicle mechanics.
 - Designed NPC AI systems using A* pathfinding and finite state machines that scaled adaptively across varied map layouts and game genres.

PERSONAL PROJECTS

- **Game Boy Color Emulator** *C, System Architecture* 2023
 - Implemented a cycle-accurate Sharp LR35902 CPU emulator in C/C++, correctly handling the full instruction set and memory-mapped I/O.
 - Passed the Blargg CPU instruction test suite in full, validating correctness against a standardized benchmark covering edge cases in real Game Boy Color ROM execution.
- **Stock Price Prediction via Time Series Analysis** *Python, ARIMA, SARIMA, Pandas, NumPy, 2025–2026*
Streamlit, Plotly
 - Built a modular Streamlit forecasting app implementing 5 models (ARIMA, SARIMA, Seasonal Naive, Moving Average, Historic Mean) on AAPL, META, and MSFT stock data, achieving 98.4% directional accuracy on a held-out test set.
 - Automated ARIMA/SARIMA hyperparameter selection (p, d, q) from ACF/PACF analysis and ADF stationarity tests, eliminating hardcoded parameters and enabling data-driven model configuration.
- **Open Source Contributions** *C, R, Data Visualization, Embedded Systems* Oct 2025 – Ongoing
 - Implemented broadcasting for tensor `matmul` on 3D and 4D matrices in [cTensor](#) (a C tensor library), extending the operator to handle arbitrary batch dimensions and writing a comprehensive test suite covering edge cases across dimension combinations; PR merged with 819 additions.
 - Traced a root-cause compiler-side bug in [animint2](#), an R interactive data visualization library: identified that `compute_domains()` fails for `geom_bar` because `built_data` exposes raw columns (x, y) while the domain lookup expects `xmin/xmax/ymin/ymax`; added a regression test and a diagnostic warning surfacing the missing columns.

EDUCATION

- **Motilal Nehru National Institute of Technology (MNNIT)** *Prayagraj, India* Expected May 2028
Bachelor of Technology in Computer Science & Engineering
- **Indian School Muscat** *Muscat, Oman* Graduated Apr 2024
Higher Secondary Examination (CBSE) – Scored 95.4%

ACHIEVEMENTS & AWARDS

Chairman's Award (2024): Awarded for achieving the highest grade (A1) in all subjects in the national CBSE board examinations.

Principal's Award for Academic Excellence (2024): Recognized for outstanding overall scholastic performance in Class 12.